Placement & Salary Prediction System

with AI-Based Insights

Machine Learning Based Career Prediction System

Team 6:

- B23CS070
- B23CS083
- B23CS095
- B23CS107
- B23CS120
- B24CS203L

College Name: Kakatiya Institute Of Technology & Science

Problem Statement

No Placement Clarity

Students lack data-driven insights into their placement readiness and chances.

No Salary Prediction

No tool exists to predict expected salary based on academic and skill profile.

Limited Career Guidance

Existing systems offer generic advice without personalized improvement plans.

Non-Intelligent Systems

Current platforms are static, non-interactive, and not powered by ML models.

Objectives

What this system aims to achieve

1. **Predict Placement**
   Use ML models to predict placement likelihood

2. **Estimate Salary**
   Forecast expected salary range for each student

3. **Classify Performance**
   Categorize students: Beginner, Intermediate, Advanced

4. **Personalized Tips**
   Generate tailored suggestions for improvement

5. **Web Application**
   Build an interactive, user-friendly web platform

Career Readiness Platform

Key Features

Single Input System

Enter details once — no repeated data entry

Combined Prediction

Placement + Salary in one unified result

Performance Level

Beginner → Intermediate → Advanced classification

Smart Suggestions

AI-driven personalized improvement tips

Analytics Dashboard

Graphs, insights, and visual trends

Technology Stack

Tools & technologies powering the system

Frontend	HTML5 · CSS3 · JavaScript
Backend	Flask (Python)
Machine Learning	Logistic Regression · Random Forest
Database	SQLite
Visualization	Chart.js

System Workflow

1

User Input

Academic & skill data entered via web form

2

Backend Processing

Flask server receives and processes the data

3

Preprocessing

Feature scaling and data normalization applied

4

ML Prediction

Models generate placement & salary predictions

5

Results & Insights

Dashboard displays predictions, graphs, and tips

End-to-end pipeline: From raw student data → ML processing → Actionable career insights

Machine Learning Model

ML-Powered Placement & Salary Prediction

Dataset

- 10,000 student records
- Features: CGPA, internships, projects,
- aptitude, soft skills, marks
- Output 1: Placement (Yes / No)
- Output 2: Salary prediction (₹)

Preprocessing

- Removed duplicates & missing values
- Encoded categoricals (Yes/No → 1/0)
- Feature scaling via StandardScaler
- Train / Test split (80:20)

Models Used

Classification:

- Logistic Regression (Best ~80%)
- Decision Tree · Random Forest

Regression:

- Linear Regression (Best R^2 ~0.72)
- Random Forest Regressor

Model Selection

- Accuracy for classification models
- R^2 & MAE for regression models
- Best classifier: Logistic Regression
- Best regressor: Linear Regression

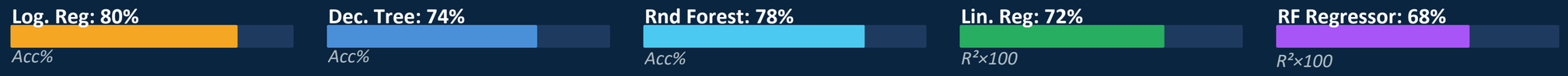
Output

- Placement: Yes / No + probability %
- Salary estimate in LPA (₹)
- Performance level classification:
  Beginner · Intermediate · Advanced
- Personalized improvement tips

Deployment

- Models saved as .pkl (Pickle files)
- Loaded at runtime by Flask backend
- RESTful API endpoints for prediction
- Deployed on Render cloud platform

Model Performance:



Live Application
Placement & Salary Prediction System
(Deployed)

https://placement-prediction-xs3j.onrender.com

Output & Results

What the system delivers to the student

Placement Prediction

Yes/No with probability score

Salary Prediction

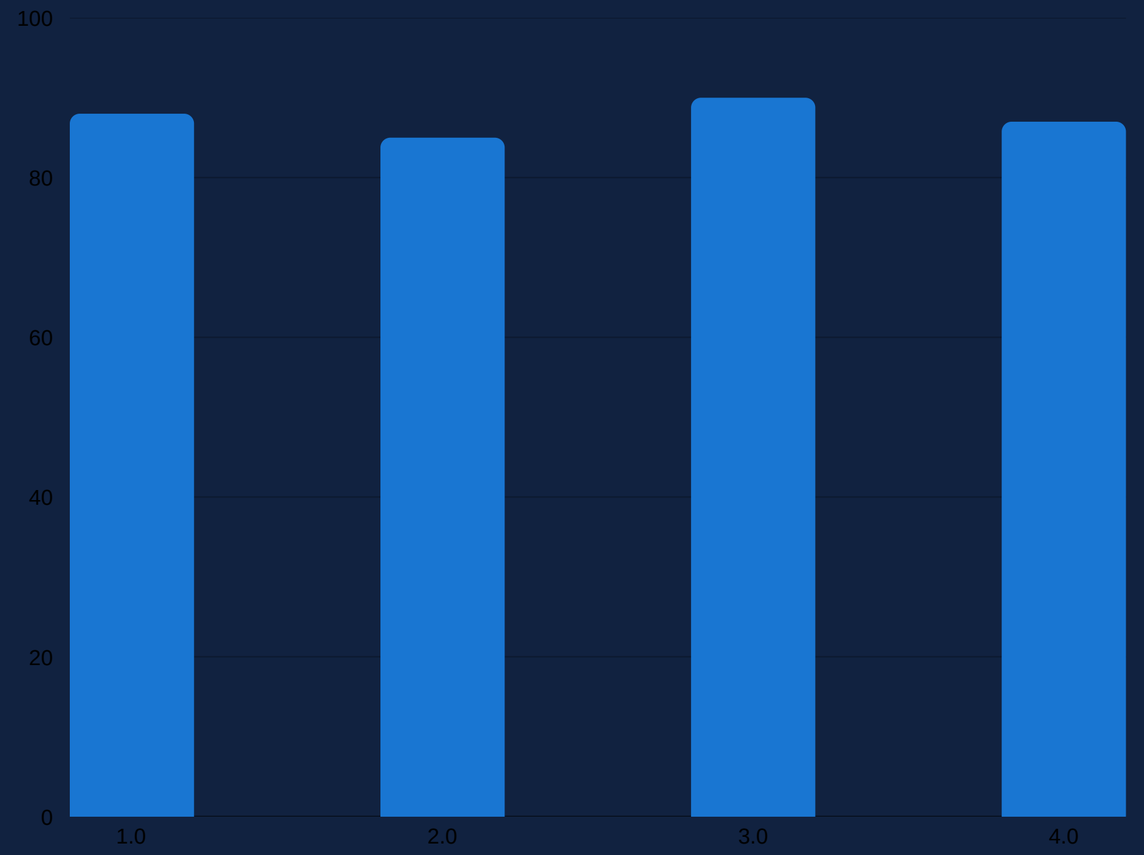
Expected salary estimate (LPA)

Performance Level

Beginner / Intermediate / Advanced

Smart Suggestions

Skills to improve, areas to focus



Conclusion & Future Scope

Conclusion

- Empowers students to evaluate career readiness objectively
- Provides data-driven, personalized placement insights
- Enhances decision-making through ML-powered predictions

Future Scope

- Resume-based automated prediction system
- AI career guidance chatbot integration
- Mobile application (Android & iOS)
- Live integration with job platforms (LinkedIn, Naukri)

Resume AI

AI Chatbot

Mobile App

Job Platforms

Building smarter careers through AI-powered insights.

Thank You.